
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Despite their impressive performance, generative image models trained on large-scale datasets frequently fail to produce images with seemingly simple concepts—e.g., human hands or objects appearing in groups of four—that are reasonably expected to appear in the training data. Because these failure modes have largely been documented anecdotally, it remains unclear whether they reflect idiosyncratic anomalies or more structural limitations of these models. This paper makes that question tractable, and its central contribution is a theoretically grounded, unsupervised framework for systematically identifying and quantifying “conceptual blindspots”—concepts present in the training data but absent or misrepresented in a model’s generations—by measuring a model’s conceptual fidelity with respect to the underlying data-generating process. Concretely, we deliver three contributions. *First*, we formalize conceptual blindspots through an energy-based measure of concept prevalence, moving the diagnosis of generative failures beyond anecdote. *Second*, we operationalize this formalization into a scalable pipeline that uses sparse autoencoders (SAEs) to extract interpretable concept embeddings and quantitatively compare concept prevalence between real and generated images; to power it, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—enabling fine-grained analysis of conceptual disparities. *Third*, we release an interactive exploratory tool and, using it, apply our framework to four popular generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), surfacing specific suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees), and—at the individual datapoint level—isolating memorization artifacts where models reproduce highly specific visual templates seen during training. Together, these contributions show that SAEs provide a principled and practical framework for uncovering conceptual blindspots in generative image models.

1 Introduction

Generative image models trained on large scale datasets have achieved unprecedented capabilities, allowing their use in applications both within the vision domain [1–9] and well beyond that [9–15]. Despite this success, several qualitative [16–18] and quantitative studies [19–20] have shown that, at times, models can struggle to generate images with relatively simple concepts, e.g., human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. In fact, when prompted to generate images containing such concepts, models tend to produce outputs with related structures, but not precisely the ground truth concept—



Figure 1: Our core diagnostic in action. The energy difference $\delta(k)$ quantifies a model’s tendency to over- or under-generate a concept c_k compared to its natural-data frequency: we deem concepts with $\delta(k) < 0.1$ **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ **exaggerated conceptual blindspots**. The depicted images, generated by four popular generative image models, show examples of images with conceptual blindspots, as well as aligned concepts—the kind of fine-grained, model-comparable evidence our framework produces automatically.

e.g., producing hands with six fingers. These failure modes, which we call “conceptual blindspots”¹ can be unintuitive, since one may reasonably expect models have had enough exposure to demonstrations accurately detailing such concepts. The key question we take up in this paper is whether such failures are merely intriguing quirks of a few specific concepts, or whether they are symptomatic of a more systematic phenomenon in which, across a broad spectrum of concepts, models are consistently either unlikely or overly likely to produce images containing them. Answering this at scale is precisely what our framework is designed to do.

No existing approach for evaluating generative image models answers this question, which is precisely the gap our framework fills [28, 29]. Specifically, existing approaches generally rely on coarse-grained measures that are meant to assess image realism, e.g., FID [30], and hence do not capture distributional failures. Methods like CLIPScore evaluate generation diversity or distribution-coverage statistics [31–33], hence offering partial insights to our question, but not at the granularity of fine-grained features or concepts [34, 35], making it difficult to identify conceptual blindspots. Finally, qualitative analyses for evaluating generative models, such as participant surveys [5, 17, 36–38] or open-ended exploration [39], can identify failures in models’ ability to capture certain concepts, but do not offer a scalable methodology that can be easily repeated across model families and used for their comparison. In short, what is missing—and what we contribute—is an automated, fine-grained, and repeatable way to pinpoint *which* concepts a model over- or under-generates.

This work. Motivated by the above, our central thesis is that identifying and analyzing conceptual blindspots in a generative image model requires a methodology that, in an automated and unsupervised manner, can elicit concepts in the data distribution whose odds of generation differ between the true data-generating process and the trained model. We develop exactly such a methodology, and demonstrate its reach across four widely used models. To make our contributions concrete, we organize the paper around the following three, each mapped to the section that delivers it:

- **Contribution 1 — Formalizing conceptual blindspots (Section 3).** We introduce a systematic, theoretically grounded framework for defining and quantifying conceptual blindspots in generative image models relative to natural images. This formalization is our conceptual foundation: it replaces anecdotal or human-defined concept evaluations with a principled, energy-based measure of a model’s limitations.
- **Contribution 2 — A scalable, unsupervised method built on sparse autoencoders (Section 4).** We turn the formalization into a concrete pipeline that uses sparse autoencoders (SAEs) to extract and compare concept distributions between natural and synthesized images. As a reusable artifact underpinning this method, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such RA-SAE to date.
- **Contribution 3 — An exploratory tool and a multi-level empirical study (Sections 5 and specifically 5.1–5.4).** We release an interactive web tool that supports distribution- and datapoint-level analysis of blindspots across models, and we use it to study Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1. This study surfaces concrete instances of both suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), shown in Fig. 1.

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.

78 To support reproducibility and adoption of all three contributions, the model, code, and web tool are
79 available at a software repository.

80 2 Related Work

81 2.1 Explainability in Vision

82 Early work in explainable AI, including computer vision, focused on methods for attribution of
83 influential input regions [40–42]. However, these methods offered limited semantic information about
84 learned representations and often produced incorrect explanations [43–45]. To address these issues,
85 concept-based interpretability [46] emerged to identify semantically meaningful directions in neural
86 networks, revealing not just where they look but what concepts and structures they employ [47–49].
87 This line of work supplies the representational vocabulary our method builds on.

88 Recent work demonstrates that popular concept-based interpretability methods—ACE [44],
89 CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—essentially address the same dictionary
90 learning task under different constraints [52]. Out of these approaches, sparse autoencoders (SAEs)
91 have emerged as particularly scalable for concept-based interpretability, which is exactly why we
92 adopt them. While recent studies reveal some limitations of the original SAEs—including overly spe-
93 cific features [53], compositionality challenges [54], and limited intervention effects [55]—improved
94 SAE versions have emerged, including the archetypal SAE (RA-SAE) [56] that our second contribu-
95 tion scales up.

96 Beyond SAEs, other interpretability methods—including prompt-based probing [57], attention
97 map or activation visualizations [58, 59], and dataset-level statistics [32, 33] such as diversity or
98 distribution coverage metrics—offer only partial insights into the question we pose. Crucially, they
99 lack granularity, focusing on full images or prompts instead of fine-grained features and concepts [34
100 35]. Furthermore, they depend on subjective interpretation and do not distinguish between various
101 failure modes [60]. These methods and metrics are therefore inadequate for systematically identifying
102 feature- and concept-level weaknesses of generative image models [28]—the precise inadequacy our
103 formalization (Contribution 1) and method (Contribution 2) are designed to remove.

104 2.2 Generative Image Models

105 Diffusion-based methods have become dominant across various modalities in generative vision
106 modeling, including image [4, 61–64], video [65–72], and 3D [7, 73–75]. In the domain of image
107 generation, this lineage can be traced back to denoising diffusion probabilistic models (DDPMs) [76],
108 which were later extended to non-Markov diffusion processes with denoising diffusion implicit
109 models (DDIMs) [77].

110 The Stable Diffusion (SD) [9] model family made DDPMs highly accessible in both the research
111 and open-source communities. The original SD was followed by several subsequent versions,
112 including SD 2 [78], SD 3 [79], and SD XL [80]. Many modifications and extensions of the SD
113 architecture have emerged, enabling additional constraints for the diffusion process (e.g., style [81, 82],
114 pose [83], and identity [84, 85]) as well as different input modalities, such as image-to-image
115 generation. Other Latent Diffusion Models built on top of SD—including Kandinsky [86], PixArt [87],
116 DeepFloyd [4, 88], and FLUX [89]—have also emerged. This diversity of architectures is what
117 makes a model-agnostic diagnostic, like the one we contribute, valuable: four of these models (SD
118 1.5/2.1, PixArt, and Kandinsky) form the testbed for our empirical study.

119 2.3 Datasets for Generative Image Models

120 The recent success of generative vision models is largely attributed to the abundance of computational
121 resources and large-scale internet datasets [90–92]. Specifically, LAION-5B [93] has played a key
122 role in the training of open-source text-to-image models, including SD and its derivatives. This
123 dataset, scraped from Common Crawl [94], contains over 5B image-caption pairs, 2.3B of which
124 are in English. Other prominent datasets include COYO-700M [95] and Conceptual Captions [96],
125 with 700M and 3M image-caption pairs, respectively. We treat LAION-5B as a proxy for the
126 data-generating process our framework compares models against.

As LAION-5B gained popularity, concerns grew over its biases [97-101]. Despite filtering attempts, harmful content persisted [97-99], including NSFW material [100] and hundreds of CSAM instances [101], prompting its temporary removal from official channels. The dataset also suffers from low-quality images [102] and internet-style captions (e.g., product descriptions) that misalign with how users prompt trained models [103]. Such dataset-level biases are one plausible origin of the blindspots our study uncovers, and our long-tail analysis (Section 5.6) makes this link explicit.

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning seeks to find sparse representations of input data, where each sample can be reconstructed using a linear combination of few dictionary atoms [104-106]. Built upon compressed sensing theory [107-108], the field evolved from early vector quantization methods [109] to sophisticated approaches including Non-negative Matrix Factorization [110-111], Sparse PCA [112], and K-SVD [113]. Recent advances include online methods [114], structured sparsity [115], and theoretical guarantees [116-117], alongside growing connections to deep learning [118-119]. This sparse-coding machinery is what our second contribution repurposes to define concepts without supervision.

Advances in sparse coding have also been leveraged to study the emergence of high-level concepts inside diffusion models [120]. Prior to diffusion models, concept-grounded interpretability was deployed to earlier generative architectures through concept-bottleneck models, which require human intervention at training time [121], and post-hoc detectors that retrofit concept supervision [122]. Both of these approaches, however, require human-defined concepts and hence inherently miss broader trends that the user does not explicitly register—a supervision bottleneck our unsupervised method removes.

Aggregate metrics (e.g., precision, recall, density, and coverage [35, 123]) and latent density scores, which predict sample quality based on the model’s latent space [124], have emerged to evaluate generative image model capabilities. While these effectively uncover distributional gaps, they offer little insight into the specific concepts that are under- or over-represented—exactly the granularity our diagnostic provides.

Together, these limitations motivate the scalable, unsupervised framework we contribute: one that can systematically identify and quantify concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

This section delivers our first contribution: a formalization of *conceptual blindspots* as systematic discrepancies between the conceptual content of natural images and that of model-generated outputs. The payoff of this formalization is that it lets us derive principled, quantitative measures characterizing which concepts a generative model under- or over-represents relative to its data distribution—measures that the rest of the paper operationalizes and applies. The process is illustrated in Fig. 2. While we rely on standard assumptions in this pursuit [126-130], we emphasize that, empirically, meaningful phenomenology is elicited even when these assumptions are violated.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational simplicity, we use $p(\cdot)$ to denote both the latent density $p(c)$ and its push-forward to image space $p(x)$, where $x = G(c)$. This is justified by the invertibility of G , which induces a valid distribution over $\mathcal{X}$ via the change of variables formula. In essence, the individual dimensions of the latent space reflect the **Concepts** underlying the data-distribution $P_{\mathcal{X}}$, defined over some observation space of images $\mathcal{X}$. For example, different latents may correspond to concepts like color, shape, size, location, and so on [131-132]. We also let the data-generating process associate a text-description $t \in \mathcal{T}$ with

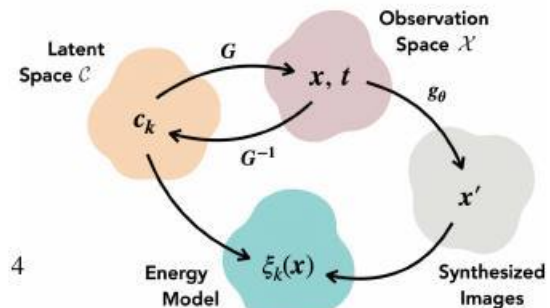


Figure 2: **Intuitive mapping of our framework (Contribution 1).** Latent concepts $c_k \in \mathcal{C}$ are

any image sampled from the data distribution, but do not explicitly model it. These text descriptions can then be used to train a text-conditioned **Generative image model** g_θ , with parameters θ , on a set of image-text pairs to map a noise signal $\eta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and a text-description of the scene t to produce images x illustrating the latter.

The crux of defining a conceptual blindspot is comparison: we must assess the probability mass a concept receives from the data-generating process against the mass it receives from the model g_θ . To this end, we take an evaluation set of natural image-text pairs $(\mathcal{D}_\mathcal{X}, \mathcal{D}_\mathcal{T})$ and define a set of synthetically generated images $\mathcal{D}'_\mathcal{X}$ using the text descriptions. To estimate the probability of occurrence of a concept, we introduce an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps images to a d -dimensional vector, where each dimension is a scalar representing the energy in the k^{th} concept, i.e., $\xi_k(x) = E(c_k)$. These estimates are feasible precisely because we assume the data-generating process is invertible. Correspondingly, the probability the data-generating process associates with the image x can then be defined as $p_k(x) \propto \exp(-\sum_k \xi_k(x))$ (where Z_ξ is the partition function), hence yielding a population-level estimate $p_k(\mathcal{D}_\mathcal{X}) = \prod_{x \in \mathcal{D}_\mathcal{X}} p_k(x)$. Using this and the sigmoid map $\sigma(\cdot)$, we now define the central quantity of our framework—the average energy difference in the k^{th} concept between the datasets $\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X}$.

Definition 2 (Energy Difference). Let $x \in \mathcal{D}_\mathcal{X}$ denote a real image sampled from the data-generating process G , and let $x' \in \mathcal{D}'_\mathcal{X}$ be a synthetic image generated by the model g_θ . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the energy difference for concept k as:

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{x'}[\xi_k(x')] - \mathbb{E}_x[\xi_k(x)]) \\ &= \frac{p_k(\mathcal{D}'_\mathcal{X})}{p_k(\mathcal{D}_\mathcal{X}) + p_k(\mathcal{D}'_\mathcal{X})}, \end{aligned} \quad (1)$$

where the expectations are taken over $\mathcal{D}'_\mathcal{X}$ and $\mathcal{D}_\mathcal{X}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{x \in \mathcal{D}} \xi_k(x))$ denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

Intuitively, then, the energy difference in the k^{th} concept captures the ratio of the probability a concept occurs in a set of observations (here, $\mathcal{D}'_\mathcal{X}$) to that in a baseline dataset (here, $\mathcal{D}_\mathcal{X}$)—a single, interpretable number per concept that is the workhorse of our entire analysis. Building directly on this measure, we define conceptual blindspots as follows.

Definition 3 (Suppressed / Exaggerated Conceptual Blindspots). Given a generative image model g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

To summarize the formalization: a conceptual blindspot is a concept whose likelihood of occurrence in generated images deviates markedly—either through suppression or exaggeration—from its prevalence under the data-generating process. Suppressed concepts exhibit disproportionately low activation (e.g., $\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are overrepresented (e.g., $\delta(k) > \lambda_{\max}$). Throughout our analysis, we adopt threshold values of $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to isolate these regimes.

It is worth situating this definition against the familiar notion of “mode collapse” studied in past work (e.g., see Bau et al. [39]): the distinction is one of granularity. Mode collapse concerns the exaggerated or suppressed odds of generating *entire images*, whereas our framework operates at the finer resolution of specific concepts. For example, if a model fails to produce images of an object with a white background, our framework labels this concept a suppressed conceptual blindspot—a level of specificity mode-collapse analyses cannot reach.

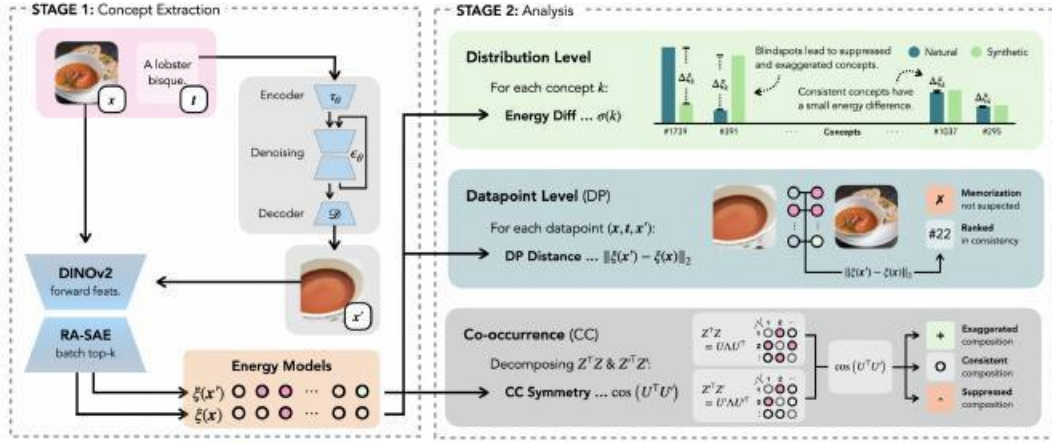


Figure 3: **Concept Extraction Pipeline (Contribution 2).** For a triplet (x, t, x') , we extract the concepts in x and x' by obtaining each image’s DINOv2 features and processing them with our RA-SAE into sparse concept embeddings, yielding energy models $\xi(x)$ and $\xi(x')$, respectively. In particular, $\xi(x) = E(x)$ holds the energy for each concept. This shared energy basis is what makes real and generated images directly comparable.

Having formalized conceptual blindspots, we now present our second contribution: a concrete, scalable pipeline that operationalizes the definition for any generative model g_θ . As per Sec. 3 this requires two ingredients: (i) a set of images sampled from g_θ that allow comparison with the ground-truth generative process, and (ii) an energy model that enables said comparison. The core methodological move of this section is to realize (ii) with sparse autoencoders, yielding an unsupervised, reusable instantiation of the energy model. Below, we use $\|\cdot\|_F$ to denote the Frobenius norm and $\|\cdot\|_0$ to denote the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix X , $X \geq 0$ implies element-wise non-negativity. For $n > 0$, we let $[n] := \{1, \dots, n\}$, and denote the i -th row of a matrix A by A_i .

From Prompts to Latent Representations. The first stage of the pipeline builds the paired data our diagnostic consumes: to identify conceptual blindspots in a model g_θ , we compare a dataset $\mathcal{D}_\mathcal{X}$ of image-caption pairs (x, t) sampled from the data-generating process G against their synthetic counterparts generated by g_θ from the same text descriptions.

Specifically, given t , we synthesize a counterpart image x' using a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$, implemented as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Sampling occurs in latent space via a reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0 is decoded via a pretrained VAE to yield the synthetic image $x' = \text{VAE}(\gamma_0)$. For the remainder of the paper, we treat g_θ as a black box that maps prompts to images: $t \mapsto x'$.

Defining the Energy Model. The second and defining stage of the pipeline instantiates the energy model. Building on prior work showing that self-supervised learning methods can invert the data-generating process and recover its underlying energy function up to linear transformations [126, 128, 133, 134], we use DINOv2 [135]—a state-of-the-art self-supervised model—as our feature extractor. Under the expectation that the number of concepts underlying the DGP exceeds the dimensionality of the model’s feature space [51, 136], we train sparse autoencoders (SAEs) on its features to isolate subspaces corresponding to these concepts [50, 56, 137–139]. The intuition, which drives our design choice, is that if the concepts underlying the generative process are represented by approximately orthogonal directions in DINOv2 (as assumed in our independence constraint in Def. 1), then an SAE should recover these concepts along individual dimensions of its latent space [136]. The activation the SAE assigns to a dimension then serves directly as our approximation of the ground-truth energy function for the concept that dimension models—closing the loop with the formalization of Contribution 1.

Formally, using $f : \mathcal{X} \rightarrow \mathbb{R}^d$ to denote our feature extraction module (i.e., the DINOv2 model), we extract features $a = f(x) \in \mathbb{R}^d$ from both natural and synthetic images from datasets $\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X}$.

Assuming the count of image-text pairs is n , we stack the real and generated features into matrices $\mathbf{A}, \mathbf{A}' \in \mathbb{R}^{n \times d}$. We then decompose each feature vector into a sparse combination of learned concept atoms using an SAE. Specifically, let $\mathbf{D} \in \mathbb{R}^{d \times K'}$ denote a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the SAE encoder that maps input features to sparse codes. Applying Ψ row-wise yields the matrix of activations $\mathbf{Z} = \Psi(\mathbf{A}) \in \mathbb{R}^{n \times K'}$, where each row $\mathbf{z}_i = \Psi(\mathbf{a}_i)$ represents the concept decomposition of an image. The decoder reconstructs features via $\mathbf{Z}\mathbf{D}^\top$, and the SAE is trained to minimize the reconstruction error subject to sparsity and non-negativity:

$$\min_{\Psi, \mathbf{D}} \|\mathbf{A} - \Psi(\mathbf{A})\mathbf{D}^\top\|_F^2 \quad \text{s.t.} \quad \Psi(\mathbf{A}) \geq 0, \|\Psi(\mathbf{A})_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

A key design decision underpins the reliability of our method: vanilla SAEs often drift toward arbitrarily oriented dictionaries, making downstream analyses highly sensitive to the random seed. To eliminate this instability—and thereby make our study reproducible and seed-independent—we employ the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary $\mathbf{D}$ to be a convex combination of training data. Specifically, we write $\mathbf{D} = \mathbf{W}\mathbf{A}$ with $\mathbf{W} \in \Omega_{K', n}$, the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \left\{ \mathbf{W} \in \mathbb{R}^{K' \times n} \mid \mathbf{W} \geq 0, \mathbf{W}\mathbf{1} = \mathbf{1} \right\}. \quad (3)$$

Thus every atom $\mathbf{D}_i$ lies in the convex hull of the data $\text{conv}(\mathbf{A})$, and any reconstruction $\mathbf{Z}\mathbf{D}^\top$ resides inside the conic hull of the data $\text{cone}(\mathbf{A})$. This guarantees that learned concepts remain faithful to the support of the data distribution [56]—a property essential to the credibility of every downstream blindspot claim. Once trained, the SAE yields a consistent set of sparse codes: $\mathbf{Z}$ for real images and $\mathbf{Z}'$ for their generated counterparts. *These codes express the same prompt-conditioned visual semantics through shared, interpretable concepts, and their activation values serve as the energy values that drive our analysis of conceptual blindspots.* In short, our method is a structured pipeline that, given a prompt and its associated real image $(t, \mathbf{x})$, emits two sparse concept vectors $(\mathbf{z}, \mathbf{z}')$ and thereby enables direct comparison of real and generated visual content in a common conceptual basis.

To recap the second contribution end to end: starting from a captioned image $(t, \mathbf{x})$, we synthesize a counterpart $\mathbf{x}'$ and map both images into a shared, sparse concept space via a vision encoder and our trained RA-SAE, yielding interpretable representations $(\mathbf{z}, \mathbf{z}')$. These representations are the foundation on which the empirical study of Section 5 rests.

5 Results

This section presents our third contribution: the exploratory tool and the multi-level empirical study it enables. Putting the method of Section 4 to work, we analyze four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—using $|\mathcal{X}| = 10,000$ image-text pairs and their corresponding generations (Fig. 4). Demonstrating the breadth of what the framework surfaces, our analysis spans three levels (Fig. 3): a **distribution-level** evaluation reveals suppressed and exaggerated concepts; a **datapoint-level** analysis surfaces failures tied to ambiguity, omission, and memorization; and a **compositional** analysis uncovers subtle distortions in concept co-occurrence geometry.

At the heart of this contribution is an interactive exploratory tool, shown in Fig. 5. The tool is a web-based interface built around a UMAP projection of concept representations, enabling visualization and comparison of concept-level energy differences. We release it publicly at a software repository, together with pre-computed energy difference data for all four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). We stress that every subsequent analysis in this paper was itself derived using this tool—it is not an afterthought but the engine of our study. Its primary functionalities, which directly support these analyses, are:



Figure 4: **Representative image-caption pairs** from LAION-5B, each with the matching images generated from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky—the paired real/generated data our framework compares.

- **Contrast different models and architectures.** For each evaluated model, the tool provides a UMAP visualization spanning all 32,000 concepts from the RA-SAE. Each scatter point represents an individual concept, color-coded by its energy difference.
- **Inspect concepts.** Each concept has a card with key statistics, representative real and generated images (x , x'), and visualized co-occurrence patterns.
- **Explore blindspots.** Beyond the UMAP and per-concept views, the tool features global rankings of suppressed and exaggerated blindspots, helping to highlight the most notable conceptual blindspots.

Rather than exhaustively studying one phenomenon, we deliberately present a breadth of high-level findings that showcase the tool’s versatility and invite broader, customizable exploration by the community.

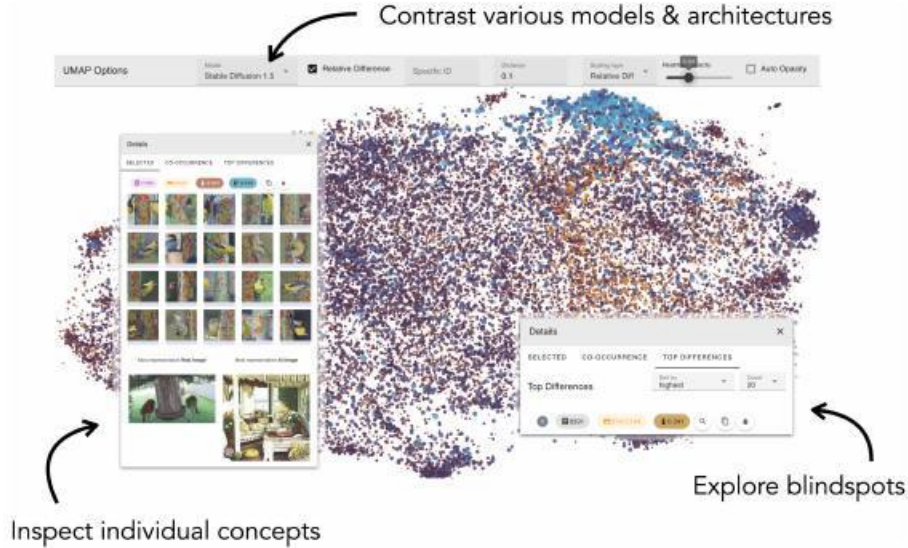


Figure 5: **Overview of the Exploratory Tool (Contribution 3).** The web interface displays a UMAP projection for each evaluated model, where each dot represents a concept, color-coded by its energy difference. Selecting a concept opens a detail panel with illustrative images, statistics, and the most representative natural and generated images x and x' . An ordered list of the concept’s co-occurrences is shown alongside global rankings of blindspots—together, the views that made every finding in this section discoverable.

5.1 Generative Image Models Cannot Produce Unseen Concepts

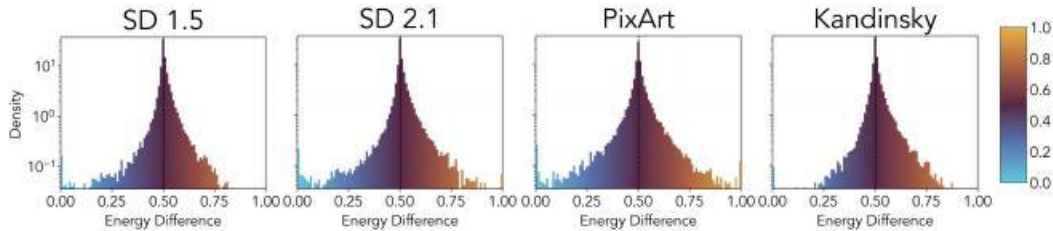


Figure 6: **Concept Energy Distribution.** Log-scale histograms of energy differences $\delta(k)$ across 32,000 concepts, comparing the natural and synthesized distributions for each evaluated model. Values left of zero mark suppressed (under-represented) concepts; values right of zero mark exaggerated (over-represented) ones. The consistent left-heavy shape across all four models is the first distribution-level evidence that blindspots are systematic, not incidental.

Our first empirical finding is a distribution-level one: generative models systematically fail to reproduce concepts they were exposed to. We establish it by analyzing the marginal energy difference $\delta(k)$ across the 32,000 concepts learned by our RA-SAE. As defined in Sec. 4, this quantity reflects

the relative prevalence of each concept in the synthesized versus natural image sets: $\delta(k) < 0.1$ signals that concept k is under-represented (suppressed) in the generated images, while $\delta(k) > 0.9$ signals over-representation (exaggerated). Fig. 6 presents the distribution of $\delta(k)$ for each of the four evaluated models. Across all models, the histograms are heavy-tailed with substantial mass at both extremes—direct evidence of systematic discrepancies in concept coverage. Crucially, the left tail—corresponding to suppressed concepts—is consistently denser and longer than the right, revealing a robust tendency toward concept suppression. This asymmetry is quantified by the negative skewness of the distributions: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

While every model exhibits both suppressed and exaggerated concepts, their specific profiles differ. PixArt, for instance, shows a wider spread, indicative of a more strongly suppressed concept distribution. The takeaway of this first analysis is that the left-skew is universal—a common tendency toward concept omission—even though its detailed shape is model-specific. The following sections dissect this behavior further.

5.2 Structure and Specificity of Conceptual Blindspots

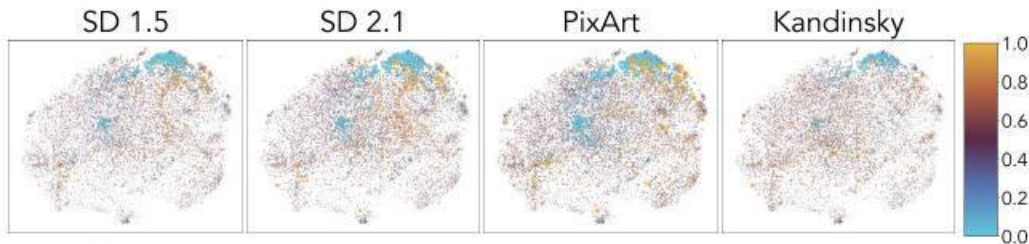


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts, colored by their energy difference $\delta(k)$ between the natural and synthesized distributions (suppressed concepts at the blue end, exaggerated at the yellow end). The emergence of coherent clusters, rather than noise, shows that blindspots are structured—a finding our framework makes visible at a glance.

Our second finding sharpens the first: blindspots are not scattered at random but are structured, and this structure is partly shared and partly model-specific. Whereas the previous section quantified marginal discrepancies in concept frequency, here we probe their global structure by embedding the full set of 32,000 concepts into two dimensions using UMAP on the sparse codes, coloring each concept by its $\delta(\cdot)$ value. As shown in Fig. 7, distinct clusters of concepts emerge across all models. These clusters often correspond to contiguous regions of conceptual blindspots, especially for suppressed (blue) concepts—strong evidence that blindspots are structured, reflecting shared biases in either the training distribution or architectural priors. To quantify how consistent these blindspot patterns are across models, we analyze both the magnitude and structure of concept-level $\delta(\cdot)$ values. Fig. 8 reports scatter plots and pairwise Pearson correlation coefficients between the $\delta(k)$ vectors of SD 1.5 and every other model. As one would expect, SD 1.5 and 2.1 correlate strongly ($r = 0.82$), reflecting their shared architectural and training pipelines. Their correlations with PixArt and Kandinsky, by contrast, are markedly lower— $r = 0.41$ and $r = 0.46$, respectively—showing that these models emphasize different regions of the conceptual space.

In sum, this analysis establishes a two-part finding: some blindspots are universally shared—likely rooted in properties of the dataset—while others are highly model-specific, emerging from idiosyncrasies in training dynamics or model capacity. This dichotomy motivates the distinction, maintained throughout the remaining sections, between blindspots shared across models and those unique to a single model.

5.3 Qualitative Examples of Blindspots

To make the preceding statistical findings concrete, we now put faces to the numbers by visualizing specific suppressed and exaggerated blindspots. In Fig. 9a we show a blindspot suppressed by *all* models: every evaluated model fails to reproduce the concept *solid white on documents*. Despite the caption explicitly referencing this concept, none of the generated images reflect the intended visual semantics, confirming that this region of the concept space is systematically under-sampled across models. Fig. 9b, by contrast, illustrates a model-specific blindspot: the concept *pan* is

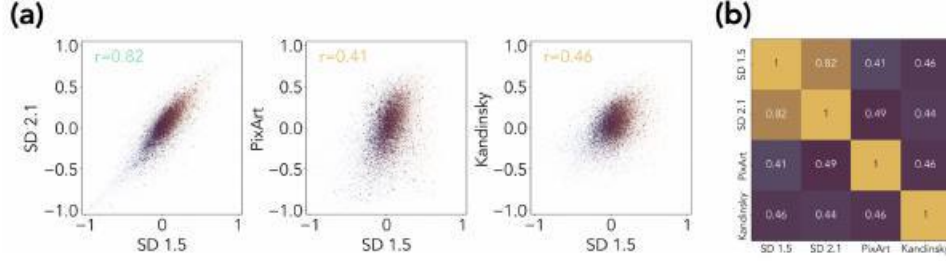


Figure 8: **● Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with Pearson correlation coefficients reported top left. The strong alignment between SD 1.5 and SD 2.1 against the weaker correlations with other architectures is our quantitative signature of model-specific blindspots. **● Correlation Matrix of Conceptual Blindspots.** Heatmap of pairwise Pearson correlation coefficients for δ between all models, summarizing how much of the blindspot structure is shared.

accurately captured by three models yet conspicuously missing from Kandinsky’s generations. This concrete example reinforces the finding of Sec. 5.2—that cross-model agreement is high in some cases and limited in others.

Turning to the opposite regime, Fig. 10 presents a case of exaggeration: the concept shadow under animal is overly emphasized in generated images. While shadows are mildly plausible, their consistent and pronounced rendering across models—set against the more nuanced and variable occurrences in natural images—points to an overactive prior. Notably, despite deliberate attempts to find concepts uniquely exaggerated by a single model, we found no clear examples, suggesting that exaggerations, unlike suppressions, are approximately universal.

Taken together, these examples concretely demonstrate the value of our energy-based diagnostic: it surfaces both shared and model-specific failure modes, and it exposes surprising limitations—such as the consistent, cross-model failure to reproduce clear or solid background elements like whitespaces on documents. That such patterns are universal raises the possibility that certain failures stem from architectural constraints or training-data biases transcending individual model idiosyncrasies.

Informative as these aggregate analyses are, they invite a sharper question, which our next contribution-driven analysis addresses: do blindspots emerge only in aggregate?



Figure 10: **● Example of an Exaggerated Conceptual Blindspot.** Four synthesized images x' generated using S_θ for the concept shadow under D'_l .

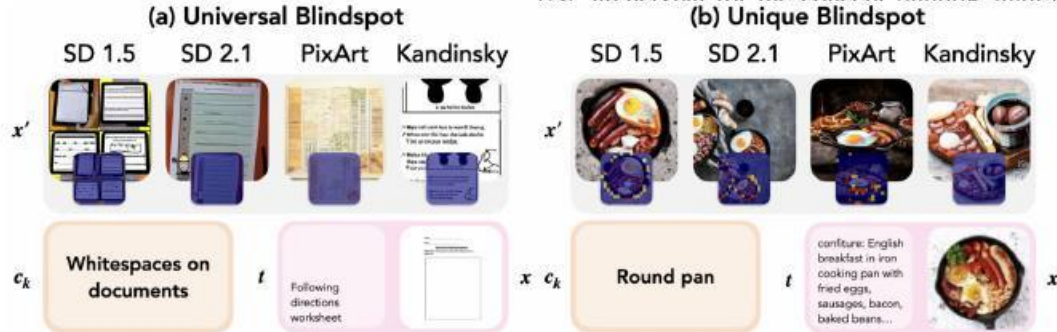


Figure 9: **● Examples of Suppressed Conceptual Blindspots.** The natural images x , representative c_k and t , shown alongside four synthesized images x' generated using S_θ . These examples make the two regimes tangible: the universal blindspot appears in all evaluated models, whereas the unique blindspot appears only in Kandinsky.

gate across many samples, or do they manifest even at the level of individual datapoints? This finer-grained perspective lets us probe the mechanisms behind blindspots directly—uncovering prompt misinterpretation, latent memorization, or both.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

Demonstrating that our tool operates at the datapoint level as well as the distribution level, we move beyond population statistics to examine individual natural vs. generated image pairs whose concept-averaged $\delta(\cdot)$ values are largest and smallest. The payoff of this datapoint-level view is that it makes model successes and failures individually inspectable, and it reveals that the failures often arise from prompt ambiguity or memorization artifacts. Concretely, Fig. 11a shows instances with near-zero difference in average $\delta(\cdot)$ values, where the generated images are conceptually indistinguishable from the original. Qualitative inspection makes clear, however, that this indistinguishability does not reflect faithful capture of prompt semantics but pure replication of memorized templates: repetitive visual structures (e.g., outlines of clothing or object arrangements) indicate the model is copying overly frequent patterns in the training data. Fig. 11b, at the other extreme, shows samples among the largest $\delta(\cdot)$ values, marking significant conceptual divergence between synthesized and natural image. Some of these discrepancies trace to underspecified or noisy captions, but others expose genuine blindspots: the prompt describes a clear concept faithfully present in $\mathcal{D}_{\mathcal{X}}$, yet the model fails to realize it in $\mathcal{D}'_{\mathcal{X}}$. The lesson is that even when language grounding is adequate, certain concepts fall outside the model’s generative abilities—a distinction only a datapoint-level diagnostic like ours can draw.



Figure 11: • **Datapoint-level Conceptual Alignment.** Two failure modes our datapoint-level analysis isolates: (a) examples with minimal energy differences, where models appear to memorize training patterns, and (b) examples with large differences, where significant concept divergences arise from prompt ambiguity or model limitations.

5.5 Analyzing Post-Training Effects

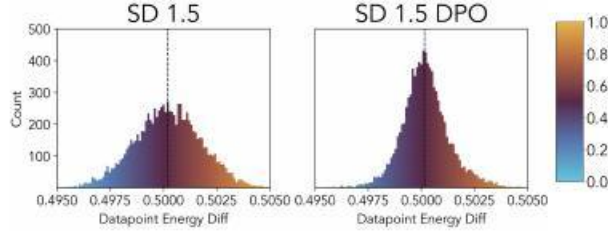


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between the synthesized and natural distributions of SD 1.5 with and without DPO. The leftward, tighter distribution under DPO is a concrete demonstration that our framework can quantify the effect of a post-training intervention.

A further contribution of our framework is that it can be pointed at questions beyond blindspot discovery; we illustrate this by using it to quantify a post-training intervention. Post-training protocols such as safety fine-tuning have been argued to reduce the diversity of model generations [140]. Leveraging our pipeline’s ability to isolate differences between a model’s generations and the ground-truth DGP, we apply it to DPO—a popular safety fine-tuning protocol [141]. Specifically, we compare two variants of SD 1.5, one trained with DPO and one without. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of the difference between their internal concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 presents a histogram of these datapoint-wise energy differences. The DPO-enhanced model exhibits both a lower median and a narrower spread, indicating that its generated concepts align more consistently with the ground-truth DGP. The natural reading is that DPO regularizes the model’s concept distribution, nudging outputs to more closely reflect the semantic content of the inputs seen during training. Our analysis does not disentangle the specific inductive biases DPO introduces, but it does provide empirical evidence that its human-preference objective indirectly promotes a better match with the training distribution—reducing *both over- and under-activation* of individual concepts relative to the natural baseline. The broader point, and the reason we include this analysis, is that our pipeline characterizes post-training interventions not merely by output quality but by how they reshape the conceptual geometry of a model’s output space.

5.6 Conceptual Misalignment as a Function of Empirical Frequency

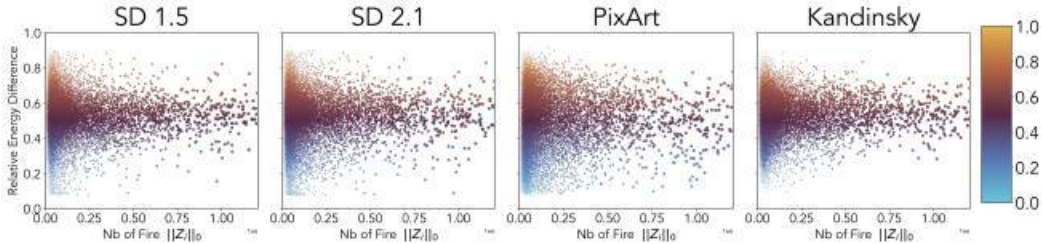


Figure 13: **Concept Fidelity Across the Frequency Spectrum.** Scatter plots relating concept frequency (x-axis) to energy difference (y-axis) across the four evaluated models; each point is a concept, sized by its activation frequency. The concentration of large errors among rare concepts is our evidence that blindspots are tied to the long tail of the data distribution.

Our final empirical finding addresses the origin of blindspots and, in doing so, connects them back to the dataset biases discussed in related work. We had hypothesized that conceptual blindspots are not merely architectural artifacts but may also emerge as a direct consequence of distributional peculiarities of certain features. Here we test this hypothesis by asking whether concepts that are rarely activated in natural images—those in the long tail of the data distribution—become blindspots in generative models. Specifically, we process the natural dataset $\mathcal{D}_X$ through the trained SAE and compute, for each concept k , its empirical frequency $\|Z_{:,k}\|_0$, the count of activations of concept k across all images. We then correlate this frequency with the absolute energy difference observed across generated outputs. Fig. 13 visualizes this relationship for all evaluated models. The finding is

clear: high-frequency concepts show low energy discrepancies, whereas rare concepts—especially suppressed ones ($\delta(k) < 0.5$)—exhibit large alignment errors. This confirms that many blindspots stem not from randomness or model quirks but from systematic effects tied to long-tail concept distributions—and it implies that mitigation may require not only architectural changes but also strategies like data reweighting or augmentation.

6 Discussion

Stepping back, this paper contributes a formalization of conceptual blindspots (Contribution 1), a scalable unsupervised method and open-sourced RA-SAE that operationalize it (Contribution 2), and an interactive tool with a multi-level empirical study across four popular generative image models (Contribution 3). The study reveals numerous conceptual blindspots, yet it only scratches the surface: each individual finding could warrant its own dedicated investigation. Rather than pursuing any single one in depth, we have deliberately used them to showcase the versatility of the method and tool. Out of the box, these enable systematic identification of concepts models struggle to generate, detection of memorization artifacts, discovery of datapoints with insufficient captions, quantification of post-training effects, and characterization of conceptual shifts across model architectures. In this sense our central contribution is a reusable framework, and we intend it to open space for follow-up work that extends the depth of analysis, the scope of evaluated architectures, and the inquiry into root causes of conceptual blindspots. One natural extension is to explore hierarchical concept representations in RA-SAE for finer-grained analysis.

Limitations. To keep the contributions honest, we note two limitations. First, because our concept extraction relies on DINOv2 and RA-SAE, the framework can only surface concepts these models represent; concepts poorly captured by them escape our analysis. Second, while our sample size of 10,000 images is substantial, it may not fully capture the long tail of rare concepts, concept co-occurrence, or other compositional statistics. Neither limitation is intrinsic to the formalization of Contribution 1—both point to concrete avenues (stronger encoders, larger evaluation sets) for strengthening the method.

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Appendix

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A Experimental Setup

To make the empirical study of Contribution 3 fully reproducible, this section details the experimental setup for our analysis of four popular generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or its subsets/derivatives. The code is available at a software repository.

A.1 Observation Space

The observation space—our proxy for the data-generating process—is constructed by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our domain of natural images. Because of concerns with CSAM and other unsafe content, the original data release is no longer available; we therefore use a substitute release of a subset with additional filtering of unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure is threefold: (1) load the full LAION dataset using the Hugging Face `datasets` library, (2) validate proper URL structure and resource availability via HTTP HEAD requests, and (3) sample randomly with replacement until reaching the target of 10,000 valid samples. This procedure yields D_G as a set of (x, t) tuples and ensures our observation space contains accessible image-text pairs, despite a large portion of the original image URLs having become unavailable since release. Additional examples of synthesized images appear in Appendix H

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} in one-to-one correspondence with D_G , yielding the triplets (x, x', t) our diagnostic consumes. Concretely, given the 10,000 image-text pairs (x, t) from D_G , we use each caption t to synthesize a counterpart image x' with each generative model g_θ .

Synthesis follows the standard text-to-image generation pipeline for each architecture, implemented with the Hugging Face `diffusers` library and models loaded at mixed precision (fp16). All synthetic images are generated at 512×512 pixel resolution with default parameters.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 (used in the analysis in Sec. 5.5) follows the baseline SD 1.5 implementation, but replaces the UNet component with a DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1 For the distribution-level finding, we compute energy differences $\delta(\cdot)$ across all 32,000 concepts for each evaluated model, applying a sigmoid transformation with temperature $T = 0.8$ during normalization. The resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

Section 5.2 To reveal the structure of blindspots, we embed the complete set of 32,000 concepts into two dimensions using UMAP on the sparse concept codes, coloring each point (a concept) by its energy difference $\delta(\cdot)$ to emphasize both suppressed and exaggerated blindspots. To quantify cross-model consistency, we compute pairwise Pearson correlation coefficients between the $\delta(\cdot)$ vectors of all model pairs, producing both scatter plots and correlation matrices. Together these expose whether blindspots cluster in conceptual space and separate model-specific from universal patterns.

Section 5.3 For the qualitative examples, we rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually examine the extrema of both regimes: concepts with $\delta(\cdot) < 0.1$ for suppressed blindspots and $\delta(\cdot) > 0.9$ for exaggerated ones. The presented examples are manually annotated with textual concept descriptions obtained by inspecting their most activating images and spatial attention patterns. Ongoing efforts to automate this concept interpretation are outlined in Appendix D.

Section 5.5 To quantify the DPO intervention, we compare SD 1.5 with and without DPO as follows: for each image pair (x, x') we compute the L2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, applying a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences before taking their mean. This yields datapoint-wise energy differences measuring how far each generated image deviates from its natural counterpart in concept space, which we visualize as overlapping histograms contrasting the two model variants.

Section 5.6 To relate blindspots to concept frequency, we count for each concept c_k its empirical frequency $\|Z_{:,i}\|_0$ (the number of non-zero activations across the natural dataset) and apply a sigmoid normalization with temperature $T = 0.4$ to the energy differences $\delta(\cdot)$. The relationship is visualized as scatter plots with empirical concept frequency on the x-axis and sigmoid-transformed energy difference on the y-axis; point sizes are proportional to activation frequency and point colors to the magnitude of the energy differences.

A.4 • Datapoint Level Analysis

Section 5.4 For the datapoint-level analysis, we compute for each image pair (x, x') the L2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, yielding a scalar measure of conceptual divergence per pair. Ranking samples by this value places potential memorization artifacts at the minimal-divergence end and significant conceptual failures at the maximal-divergence end, enabling qualitative inspection of specific failure modes.

A.5 • Co-occurrence Analysis

Appendix F For the compositional analysis, we study concept co-occurrence in both the natural and synthesized data D_G and D_{go} through the co-activation matrix $Z^T Z$, which holds pair-wise correlations in concept usage. We examine the dominant conceptual directions via eigendecomposition and assess the alignment between natural and synthetic co-occurrence structures using cosine similarity heatmaps between the top-100 eigenvectors of each co-occurrence matrix. In these 100×100 similarity matrices, perfect diagonal alignment would indicate identical principal concept axes, while off-diagonal patterns reveal rotations and mismatches in compositional geometry.

B Custom RA-SAE Training Details

This section documents the training of our custom archetypal SAE (RA-SAE) [56]—the open-sourced artifact underpinning Contribution 2 and the largest such SAE to date. The model is available at a software repository.

Dataset. The auto-encoder is trained on the complete ImageNet-1k training split (≈ 1.28 M RGB images). Each image is converted to 261 visual tokens using DINOv2 [135], and tokens are fed to the SAE without class or position embeddings, giving a total of $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$ training tokens.

Dictionary. The dictionary spans 32,000 concept dimensions. Sparsity is enforced by top- k masking with $k = 5$: activations outside the largest five per input are set to 0. Weights are initialized with Xavier/Glorot, and training is conducted at mixed precision (fp16), with the last ten epochs at full precision.

Optimizer and Schedule. We train for 50 epochs with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and weight decay 10^{-5} . A linear warm-up over the first 5% of steps is followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The objective combines an MSE loss with an auxiliary term ($\lambda = 10^{-5}$) that penalizes activations never entering the top- k set.

1011 C Computational Resources

1012 For transparency and to aid reproduction, this section summarizes the GPU resources used for the
1013 training and experiments behind our contributions. In total, we used approximately 202 GPU-hours
1014 on NVIDIA H100s and H200s.

1015 **RA-SAE.** Trained for approximately 24 GPU-hours on three NVIDIA H100s.

1016 **Synthesized Images.** Generating the full $D_{\theta\theta}$ (see Appendix A) took roughly 5 hours per generator
1017 when distributed across four NVIDIA H200 GPUs. With five generators, this totaled approximately
1018 100 GPU-hours on a NVIDIA H200.

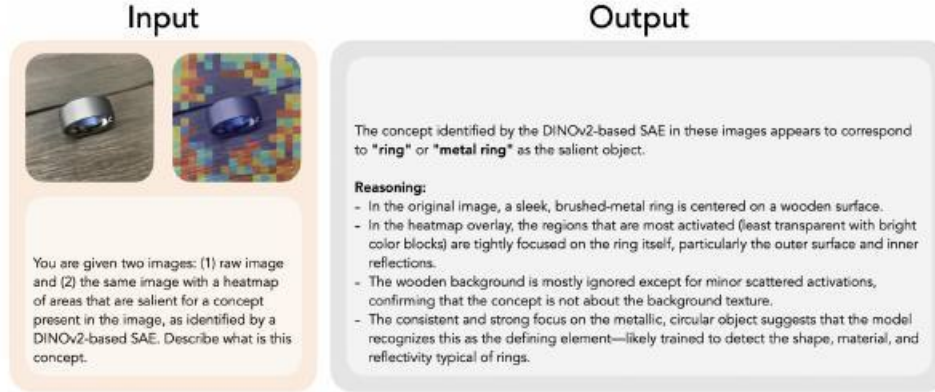
1019 **● Distribution-Level Analysis.** Extracting energy differences at the distribution level took about 3
1020 hours per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

1021 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
1022 datapoint, and ranking datapoints per concept also took approximately 3 hours per generator on one
1023 H200 GPU (total ~ 15 GPU-hours).

1024 D Autointerpretability

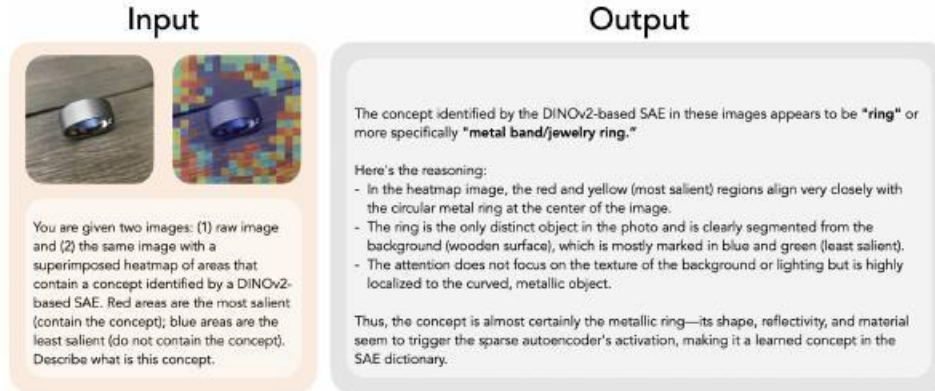
1025 Looking toward automating a step that Contribution 3 currently performs manually, we qualitatively
 1026 evaluated whether Vision-Language Models (VLMs) can describe the concepts our RA-SAE identifies.
 1027 Concretely, we queried ChatGPT 4o via the OpenAI API with various prompts and formats of an
 1028 image x'' in which the given concept was salient, recording the VLM’s description. This section
 1029 reports what we learned. For evaluation we let the VLM provide its full reasoning; a scalable
 1030 deployment would simply modify the prompt to request only the description, filtering the reasoning
 1031 out.

1032 First, we provided the VLM with two images: a raw version of x'' and x'' with a heatmap superim-
 1033 posed to localize the intensity of the concept:



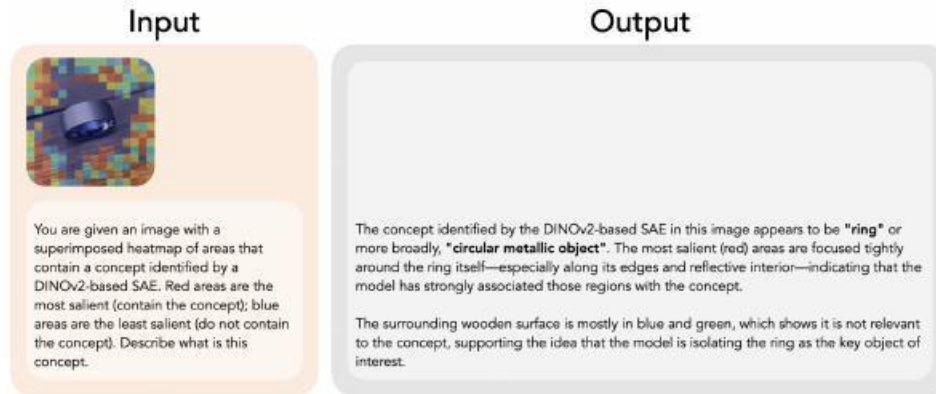
1034 As seen in the example, the VLM did not interpret the heatmap visualization correctly. Instead of
 1035 describing the wood texture in the background—the actual salient region—it described the ring in the
 1036 foreground, which received no attention from the RA-SAE.

1037 To determine whether this was a prompting issue, we tested the same scenario again—this time using
 1038 an expanded prompt that explained the heatmap in greater detail:



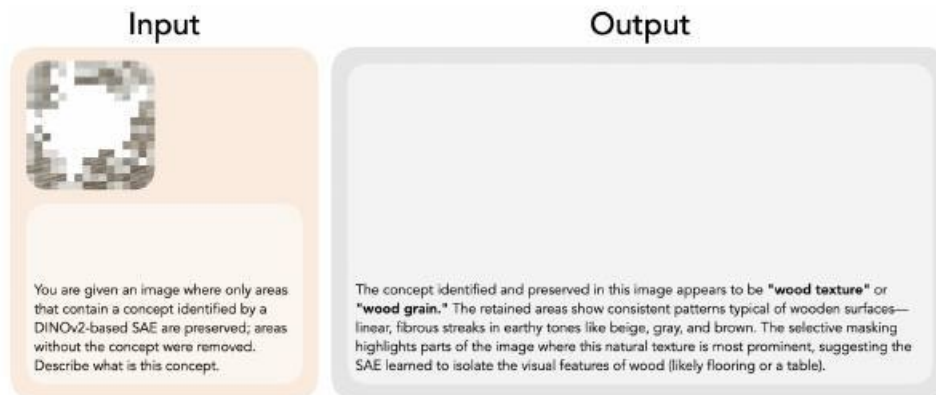
1039 This prompt modification did not resolve the issue, as the VLM still identified the ring as the
 1040 prominent concept in the image. Several other prompting techniques, including those describing the
 1041 heatmap in even greater depth (e.g., by including the code used for its generation), were explored
 1042 with similar results.

1043 We next tested providing the VLM only with the heatmap-superimposed x'' and a similar prompt
1044 explaining how to interpret the heatmap:



1045 Once again, the result failed to identify the wooden background as the prominent concept in the image
1046 and instead focused on the ring. As in the previous case, we experimented with various prompting
1047 strategies using this image format, but the results remained largely unchanged.

1048 These observations led us to conclude that the heatmap alone cannot reliably steer the VLM's
1049 attention to the salient regions: the presence of less relevant regions distracts the model. To remove
1050 this distraction, we replaced the heatmap-superimposed x'' with an alpha-masked version of x'' :



1051 As shown in the example, the VLM was now able to correctly identify wood texture as the dominant
1052 concept. We observed consistent success across a range of concepts and images.

1053 The x'' formatting and prompt shown above gave the most reliable results in our qualitative evaluation.
1054 We stress, however, that this evaluation is limited by its qualitative nature (there are no ground-truth
1055 annotations) and its focus on a single VLM. We offer it as a starting point that future work on the
1056 autointerpretability of SAE concepts can build on and expand.

E Additional Results: Model-Specific Blindspots

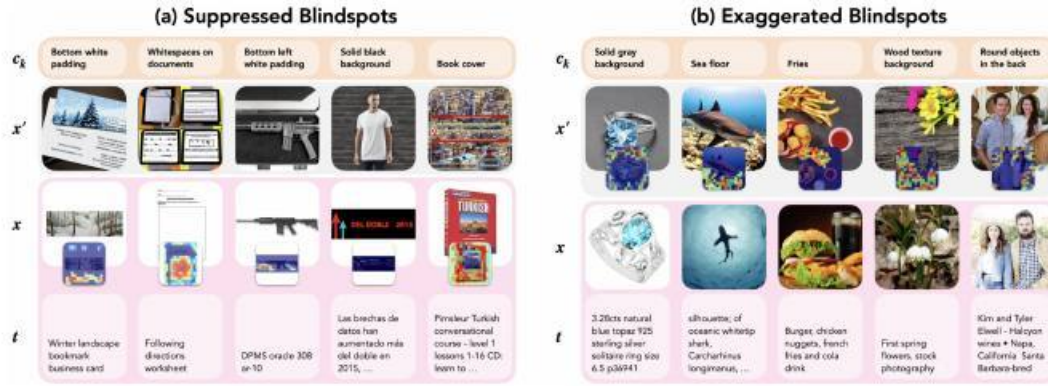


Figure 14: Model-specific conceptual blindspots surfaced by our tool for **Stable Diffusion 1.5**. For each concept we show the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots)—whichever has the highest absolute activation—with the concept’s spatial heatmap superimposed.

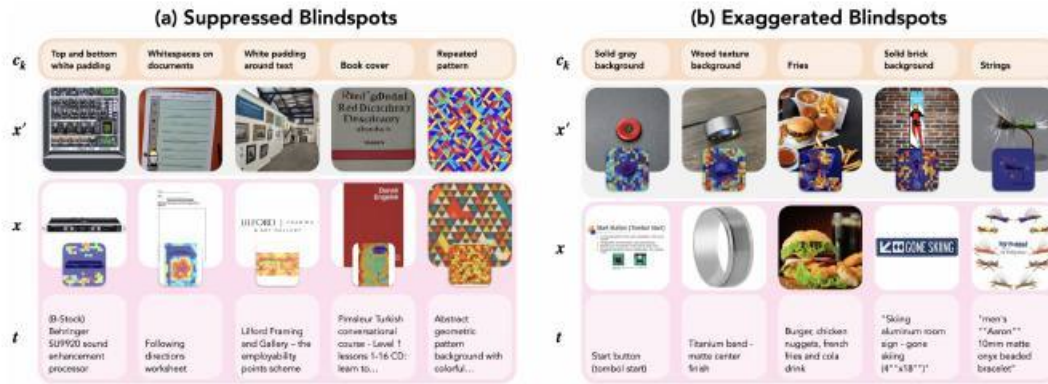


Figure 15: Model-specific conceptual blindspots surfaced by our tool for **Stable Diffusion 2.1**. For each concept we show the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots)—whichever has the highest absolute activation—with the concept’s spatial heatmap superimposed.

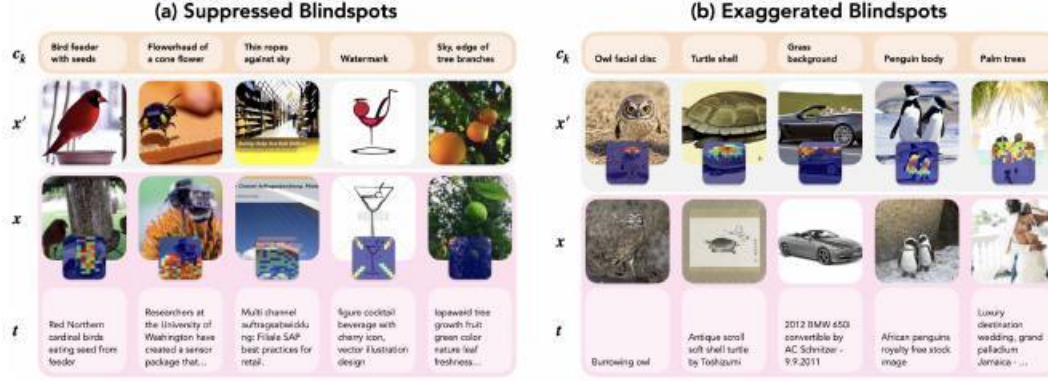


Figure 16: Model-specific conceptual blindspots surfaced by our tool for **Kandinsky**. For each concept we show the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots)—whichever has the highest absolute activation—with the concept’s spatial heatmap superimposed.



Figure 17: Model-specific conceptual blindspots surfaced by our tool for **PixArt**. For each concept we show the prototypical natural image (for suppressed blindspots) or synthesized image (for exaggerated blindspots)—whichever has the highest absolute activation—with the concept’s spatial heatmap superimposed.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

The analyses in the main paper centered on individual concept activations. Yet visual scenes are rarely composed of isolated concepts; they are structured by rich co-occurrence patterns that encode compositional semantics. As an additional contribution-extending analysis, we examine whether generative models capture this higher-order structure, using the co-activation matrix $Z^T Z$, which reflects pairwise correlations in concept usage.

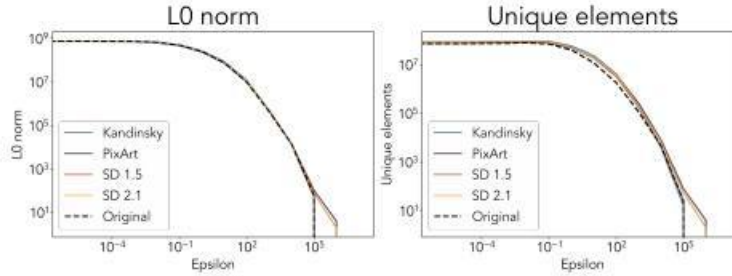


Figure 18: **Sparsity and Structural Divergence.** Left: the L_0 norm of the co-occurrence matrix $Z Z^T$ as a function of the threshold ϵ , indicating how many entries remain active in each model. Right: the number of unique entries in the synthesized distribution relative to the natural one. The key takeaway is that all evaluated models preserve global sparsity structure yet diverge in activation content.

Our first co-occurrence finding is a positive one: at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As Fig. 18 (left) shows, the ℓ_0 norm of $Z^T Z$ —thresholded at varying ϵ values—tracks closely between the natural and synthesized distributions across all models. In other words, the gross connectivity of the conceptual graph—which concepts tend to co-activate at all—is well preserved. Formally, $Z^T Z$ can be read as the adjacency matrix of a weighted, undirected graph over concepts, with edge weights reflecting co-activation strength across the dataset.

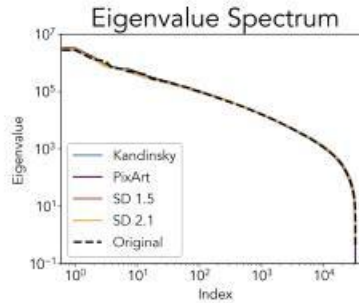


Figure 19: **Spectral Structure of Co-occurrence.** Log-log plot of the eigenvalue spectra of the co-occurrence matrices $Z Z^T$ across models and the natural distribution. That all evaluated models match the heavy-tailed decay of the natural distribution is evidence that second-order variance allocation is preserved.

The complementary—and more cautionary—finding, in Fig. 18 (right), is that the specific content of these co-activations diverges: a substantial fraction of entries in the model-generated $Z^T Z$ are not shared with the natural baseline. The capacity for compositionality is thus retained, but the identity of active pairings shifts, plausibly reflecting model-specific inductive biases or training artifacts. To probe the internal structure of these patterns, we turn to spectral analysis. Fig. 19 shows the eigenvalue spectra of the co-occurrence matrices for each model and the natural distribution. All spectra exhibit heavy-tailed, power-law-like decay, indicating that generative models preserve the overall rank structure and variance allocation across conceptual dimensions.

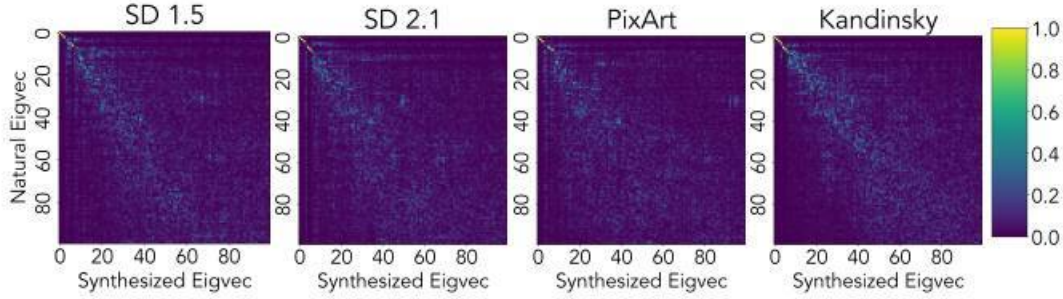


Figure 20: ● **Concept Basis Similarity.** Cosine similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . The diagonal structure indicates alignment of dominant conceptual directions, and the varying degrees of alignment across the four models are the compositional counterpart of the model-specific blindspots seen in the main paper.

1081 Finally, we examine the alignment of dominant conceptual directions via cosine similarity heatmaps
 1082 between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20).
 1083 All models show partial diagonal alignment—overlap in principal concept axes—but the off-diagonal
 1084 entries expose rotations and mismatches in higher modes, i.e., real deviations in compositional
 1085 geometry.

1086 The overall message of this section is nuanced: diffusion models approximate the global shape of
 1087 concept co-activation surprisingly well, yet deviate in subtle, structured ways under the spectral lens.
 1088 Such higher-order discrepancies may well underpin failures in generating coherent, multi-object
 1089 scenes or relational concepts—a promising direction our framework is equipped to pursue.

1090 G Additional Results: Blindspot Stress Testing

1091 To validate that the blindspots our method identifies are real rather than artifacts of the diagnostic, we
 1092 stress-test them: we gather many prompts describing each blindspot, generate images from them, and
 1093 contrast outputs from models in which the concept was flagged as a blindspot against those in which
 1094 it was not.

1095 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1096 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1097 *text-to-image model. Return these as a list of strings in Python.* We generated five images per prompt,
 1098 analyzed them with our custom RA-SAE model (see Appendix B), which ranked them by the intensity
 1099 with which the desired concept appeared, and manually reviewed all images to judge whether the
 1100 blindspot was successfully depicted.

1101 The result corroborates our method: in the following examples of suppressed concepts, while some
 1102 aspects of the target occasionally appeared (e.g., a holder or string for the bird feeder blindspot
 1103 and a round hole for the glossy DVD disc blindspot), the models generally failed to render the full
 1104 concept—exactly as our diagnostic predicted, which supports the validity of the stress test.

1105 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Stress-testing the bird feeder concept across prompts. Our method flagged this concept as a suppressed blindspot in Kandinsky, and the generations bear this out: the model cannot produce a corresponding image. SD 1.5, where the concept was not flagged, generates it successfully—a direct confirmation of our diagnostic.



Figure 22: Stress-testing the glossy DVD disc concept across prompts. Our method flagged this concept as a suppressed blindspot in SD 1.5, and the generations confirm it: the model cannot produce a corresponding image. Kandinsky, where the concept was not flagged, generates it successfully—again matching our diagnostic’s prediction.

1107 H Additional Examples: Synthesized Images



Figure 23: Additional image-caption pairs from LAION-5B, each shown with the matching images generated from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky—further illustrating the paired data our framework analyzes.

1108 I Concentration bounds for δ

1109 This section provides a theoretical guarantee for the estimator at the heart of our framework. In our
 1110 experiments we estimate $\delta(k)$ from $n = 10,000$ paired samples per concept—a modest budget that
 1111 raises the natural question of whether it suffices for reliable estimates. We answer it affirmatively by
 1112 deriving a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ via McDiarmid’s inequality [142].
 1113 The bound is tight and shows that even relatively few samples yield fast and accurate estimates of
 1114 concept bias.

1115 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1116 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1117 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1118 *Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies*

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1119 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1120 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1121 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1122 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1123 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1124 which proves the claim. □

1125 In practice, most concept scores $\delta(k)$ are sparse—concentrated near zero, with only a few reaching
 1126 values up to 10. Even for the largest observed biases, the bound guarantees that $n = 10,000$ samples
 1127 yield estimates of $\hat{\delta}_n(k)$ deviating from the true value by no more than a small ε with high probability.
 1128 This justifies our sampling strategy and confirms that accurate bias measurements—and hence the
 1129 reliability of Contribution 3’s findings—are attainable with limited data.

1130 J Monotonicity and Calibration-Free Interpretation of δ

1131 A second theoretical guarantee concerns how we rank concepts. Our aim when analyzing blindspots
 1132 is to order concepts by the severity of their generative bias, and in practice we use the score $\delta(k)$ for
 1133 this. One might worry that this score distorts the ordering relative to more direct quantities such as
 1134 the energy gap or the odds ratio. The following result dispels that worry: $\delta(k)$ is a strictly increasing
 1135 reparameterization of both, and therefore inherits their ordering exactly, so no calibration is needed
 1136 to rank concepts by $\delta(k)$.

1137 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1138 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_k}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_k}[\xi_k(\mathbf{x})],$$

1139 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1140 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1141 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1142 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1143 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1144 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1145 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1146 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1147 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1148 concepts.

1149

□

1150 In short, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by the energy gap or by
 1151 the conceptual generation odds ρ_k : all three preserve the same total ordering, and no calibration is
 1152 necessary. This underwrites the blindspot rankings surfaced by our tool.

K Stability of FID Under SAE Embeddings

This section adds a further guarantee about the SAE embedding underlying Contribution 2: we establish a quantitative relationship between the Fréchet Inception Distance (FID) computed in the original activation space (of dimension d) and the FID after applying a (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout we assume D has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We begin with a simple fact: if D is orthogonal and not overcomplete ($k = d$), then it is an isometry between A and Z , so FID is perfectly preserved. Since this case is unrealistic, we then treat the overcomplete case and show that FID can be bounded by the extreme singular values of D . We work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is just $\mathcal{W}_2^2$ specialised to Gaussians.

For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically one usually finds $\sigma_{\min}, \sigma_{\max} \approx 1$, though the proof does not rely on this. We begin with a simple lemma for the case where D is not overcomplete.

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical. $\square$

This case, however, is unrealistic, since SAEs rely on overcompleteness to extract meaningful and interpretable concepts. In the overcomplete case D is no longer orthonormal, but it can still be column-orthonormal, and we use that structure to bound the distortion via the extremal singular values of $D^\top D$.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1187 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1188 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1189 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1190 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1191 finishes the proof. $\square$

1192 The takeaway of Theorem 6 is reassuring for our method: applying a column-orthogonal overcomplete
 1193 SAE dictionary cannot distort the Fréchet Inception Distance by more than the square of its extremal
 1194 singular values. When D is nearly orthogonal—empirically we consistently found $\sigma_{\min}, \sigma_{\max} \approx 1$ —
 1195 the FID measured in the SAE feature space is essentially the canonical FID, so working in our concept
 1196 space does not compromise this standard fidelity measure. $\square$

1197